

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems(BL:3)

Assessment Tool Weightage: 10%

QUESTIONS: 4
Duration : 1 Hour

Total Marks:20

Technical Presentation

Code of Conduct:

I V. Smylika (192372115) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V. Smylika

Q1: NFA to DFA

- Define NFA and DFA
- Subset construction steps
- Challenges: state explosion
- Draw NFA and equivalent DFA from your question paper.

Q2: Mathematical Induction

- Prove $\sum i^2 = n(n+1)(2n+1)/6$
- Base case $n=1$
- Assume true for k
- Prove for $k+1$ by substitution and simplification.

Q2 Proof

- Show:
- $k(k+1)(2k+1)/6 + (k+1)^2$
- $= (k+1)(k+2)(2k+3)/6$

Q3: ϵ -NFA to NFA

- Compute ϵ -closures
- Remove ϵ -transitions
- Update final states
- Draw transition table.

Q3 Transition Table

- Create transition table without ϵ transitions from given automaton.

Q4: ϵ -Closure & DFA

- Find ϵ -closure of every state.
- Use subset construction.
- Mark start/final states.

Q4 DFA Table

- Add DFA transition table and final DFA diagram.

Conclusion

- NFA, DFA and ϵ -NFA are equivalent in power.
- DFA is used for implementation.
- Induction proves automata properties.

References

Hopcroft & Ullman
Michael Sipser
Course Notes

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/100

[Signature]